

Matrix inequalities and norms

Audenaert's two-row Schatten norm-compression conjecture

Difficulty: extreme **Importance:** broadly interesting

Problem statement

Let $N, r_1, r_2, c_1, \dots, c_N$ be positive integers and let $A_j \in \mathbb{C}^{r_1 \times c_j}, B_j \in \mathbb{C}^{r_2 \times c_j}$. For $1 \leq p < \infty$, let $\|X\|_p = (\sum_i \sigma_i(X)^p)^{1/p}$, where $\sigma_i(X)$ are the singular values. Define

$$T = \begin{bmatrix} A_1 & \cdots & A_N \\ B_1 & \cdots & B_N \end{bmatrix}, \quad C_p(T) = \begin{bmatrix} \|A_1\|_p & \cdots & \|A_N\|_p \\ \|B_1\|_p & \cdots & \|B_N\|_p \end{bmatrix}.$$

Do the following inequalities hold for every such partition?

$$\|T\|_p \geq \|C_p(T)\|_p \quad (1 \leq p \leq 2), \quad \|T\|_p \leq \|C_p(T)\|_p \quad (2 \leq p < \infty).$$

Why it matters

The question asks whether the norm of a large block matrix can be controlled sharply using only the norms of its blocks. Such compression estimates are useful when analyzing matrix approximation errors without retaining the full matrices.

References

1. K. M. R. Audenaert, *On a norm compression inequality for $2 \times N$ partitioned block matrices*, arXiv:math/0702186v2 (26 March 2007), Conjecture 1, equations (1)–(2), §§2 and 4. Published in *Linear Algebra and its Applications* 428(4) (2008). [Primary text](#).
2. T. Zhang, *Clarkson–McCarthy type inequalities, part I: ℓ_p – ℓ_p and ℓ_q – ℓ_p Schatten p -estimates*. arXiv:2410.21961v4 (28 March 2026), Conjecture 1.14 and Theorem 1.15. [Primary text](#).

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The 2026 revision still states this conjecture and supplies a weaker bound. Audenaert proves several restricted cases, including $p \geq 4$; the three-row generalization has counterexamples. Hanner's inequality, a special case, has subsequently been proved, which does not settle arbitrary two-row compression. Searches included *Audenaert norm compression conjecture 2026*, *two row Schatten compression proof*, and *norm compression counterexample 2025 2026*; no general resolution was located. Latest arXiv versions were checked for both references.